
KnowSelf

Teaching LLM Agents When to Think,
Reflect, or Seek Knowledge

Agentic Knowledgeable Self-Awareness
for LLM-Based Agents

arxiv.org/abs/2504.03553



Current Agent Training Uses 'Flood Irrigation'

Indiscriminate Knowledge Injection Hurts Efficiency

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1 Trajectory Imitation

e.g., ReAct

Blindly copies gold trajectories without assessing whether the agent can act correctly on its own.

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2 Trial-and-Error

e.g., Reflexion

Reflexively applies self-reflection at every step, even when the agent already knows the right action.

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Knowledge-Augmented

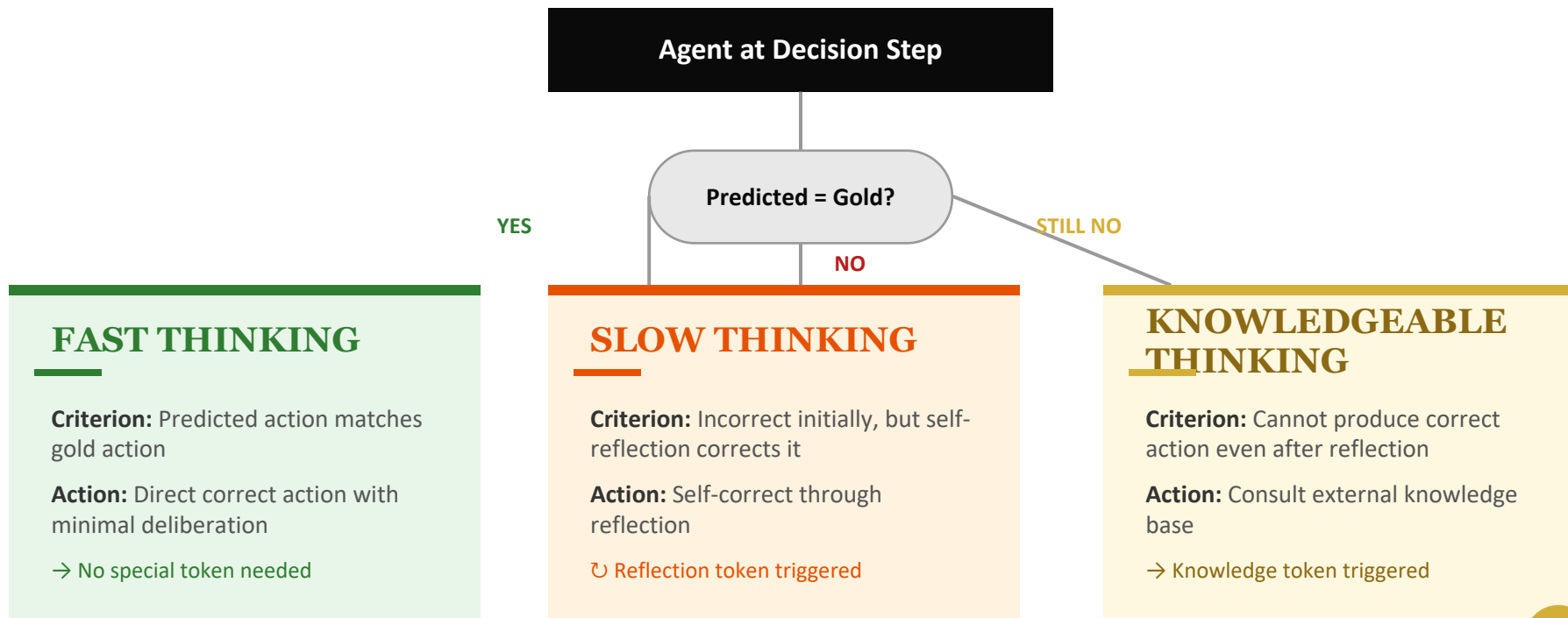
e.g., KnowAgent, WKM

Injects domain knowledge at every step regardless of situational need, creating unnecessary cost.

Humans: Situational self-awareness — know when to act, reflect, or seek help
Agents: No metacognitive regulation — all steps receive the same treatment

Three Situational Thinking Modes

Decision criteria derived from agent ability at each step



KnowSelf Framework

Data Construction → Two-Stage Training → Self-Aware Inference

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1 Self-Awareness Data Construction

Agent self-explores the environment

Heuristic judgment at each step:

- Correct → Fast Thinking
- Reflection fixes → Slow Thinking
- Needs help → Knowledgeable

Thinking

Special tokens inserted to mark each situation

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2 Two-Stage Training

Stage 1: Supervised Fine-Tuning (SFT)

- Teaches initial self-awareness patterns
- Learns to associate situations with tokens

Stage 2: Reward-based Policy
Optimization (RPO)

- Refines special token generation
- Improves situation discrimination

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3 Self-Aware Inference

Agent generates special tokens
autonomously

Token triggers appropriate behavior:

- Direct action (no token needed)
- Reflection token → self-reflection
- Knowledge token → external retrieval

Selective, efficient knowledge
utilization

KnowSelf Correctly Heats a Mug and Places a Saltshaker

Where baselines fail, KnowSelf applies the right situational thinking mode

TASK A

"Put a hot mug in cabinet"

OpenAI-O1 (fails)

Opens microwave, then incorrectly tries to take an egg instead of heating the mug

KnowSelf (Llama-8B) — succeeds

Reflection Token

Realizes the open microwave should be used to heat the mug, not to take an egg

TASK B

"Put a saltshaker in drawer"

DeepSeek-R1 (fails)

Reflects that drawer is open, tries to close it first — which fails the task

KnowSelf (Llama-8B) — succeeds

Knowledge Token

Applies knowledge: place the object in drawer without removing unrelated items

KnowSelf Achieves 84.33% on ALFWorld with Only 15% Knowledge

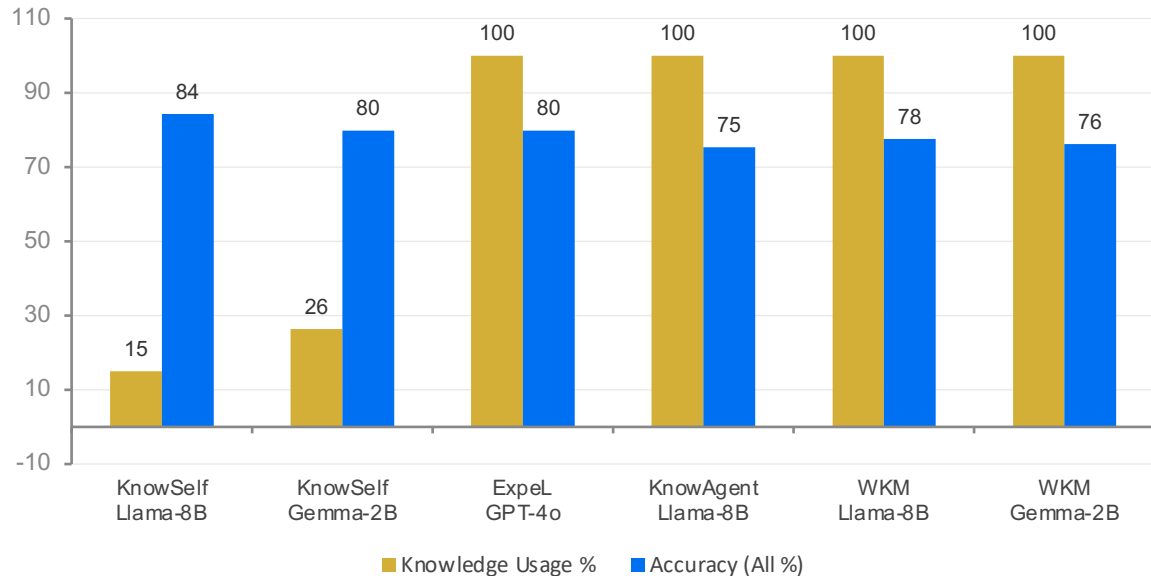
Success rates (%) across six ALFWorld task types — Know% = proportion of steps using external knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two
GPT-4o		0%	83.33	74.19	69.57	85.71	77.78	64.71
		0%	100.00	87.10	73.91	90.48	83.33	70.59
		100%	95.83	83.87	69.57	80.95	88.89	52.94
Llama-8B		0%	33.33	3.23	0.00	57.14	66.67	23.53

Key: KnowSelf (Llama-8B) outperforms GPT-4o Expel (79.85%) and all Llama-8B baselines while using only 15% knowledge. 100% knowledge methods are highlighted in red.

Minimal Knowledge, Maximum Impact

KnowSelf's efficiency advantage over full-knowledge methods



15%

Knowledge needed
for best Llama-8B result

6-

7x

More knowledge used by
competing methods, yet lower accuracy

26

%

Knowledge for Gemma-2B
still matches GPT-4o Expel

Selective knowledge use (15-26%) consistently outperforms indiscriminate injection (100%) across all model scales

Key Takeaways

Self-Awareness Is the Missing Piece in Agent Planning

- 1 Indiscriminate knowledge injection is wasteful and suboptimal — 100% knowledge methods underperform selective use
- 2 Agents can learn situational self-awareness via special tokens with lightweight heuristic labeling — no expensive human annotation needed
- 3 KnowSelf (Llama-8B) achieves 84.33%, surpassing GPT-4o ReAct (76.12%) and GPT-4o Expel (79.85%)
- 4 Only 15-26% knowledge usage needed versus 100% for competing methods — dramatic efficiency gain
- 5 The paradigm generalizes across model scales — even Gemma-2B (79.85%) matches GPT-4o Expel with KnowSelf

Source: arxiv.org/abs/2504.03553